

SooJean Han

Assistant Professor, Director of Autonomous Control for Stochastic Systems (ACSS)
School of Electrical Engineering, Korea Advanced Institute of Science & Technology (KAIST)
[soojean\(at\)kaist.ac.kr](mailto:soojean(at)kaist.ac.kr) | linkedin.com/in/soojean-han | soojean.github.io

EDUCATION

California Institute of Technology (Caltech)

Ph.D. Control and Dynamical Systems

Thesis Defense Video: [Link]

Pasadena, CA, USA

Sep. 2017 - Jan. 2023

University of California, Berkeley (UC Berkeley)

B.S. Electrical Engineering and Computer Science

B.S. Applied Mathematics

Berkeley, CA, USA

Jun. 2014 - Dec. 2016

Jun. 2014 - Dec. 2016

RECENT WORK & EMPLOYMENT HISTORY

Korea Advanced Institute of Science & Technology (KAIST).

Aug 2023 - present

Assistant Professor (School of Electrical Engineering).

Director and founder of the Autonomous Control for Stochastic Systems (ACSS) research laboratory, where we use mathematical theory and real-world applications to develop autonomous control, estimation, and decision-making algorithms, especially for stochastic systems.

Associate Editor, IEEE Robotics and Automation Letters (RA-L).

Jan 2025 - present

Joined the Editorial Board for IEEE RA-L in the area of Mechanisms, Design, and Control.

Associate Editor, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

Apr 2025 - present

Joined the Editorial Board for IROS2025 in the area of Design, Mechanisms, and Control.

California Institute of Technology (Caltech).

Feb 2023 - Aug 2023

• *Postdoctoral Scholar.*

Autonomous Robotics and Control Lab (ARCL). Performed experiments for autonomous aerial visual navigation in low-light environments (e.g., night, heavy fog), and diffusion-based motion planning. Models were trained on thermal near-shore data curated by the lab. Supported by DARPA Learning in Control (LINC) and ONR.

• *Graduate Research Assistant.*

Doyle Lab and ARCL. Performed research broadly at the intersection of control theory and artificial intelligence. Concrete applications include stochastic systems, networks and distributed sensing, and autonomous decision-making. Supported by NSF GRFP.

NASA Jet Propulsion Laboratory (JPL).

Jun 2018 - Aug 2018

Research Affiliate.

Involved with the DARPA Subterranean (SubT) Challenge as a part of Team CoSTAR. Implemented vision-based SLAM using C++ and deployed it on terrestrial rolling-walking hybrid robots custom-built by our team. Objective was to perform distributed collaborative mapping and merging using keypoint feature detection.

CAST, Caltech.

Jun 2017 - Aug 2017

Research Assistant.

Center for Autonomous Systems & Technologies (CAST). Implemented distributed control over multiple Crazyflie quadrotor platforms for synchronization and formation-flight. The combined PID and nonlinear controllers were designed in Python and C++, and deployed through ROS and VICON motion capture.

Hybrid Systems Lab. Developed and implemented obstacle detection for FaSTrack, a modular framework for autonomous path-planning. Relevant publications [here] and [here].

PUBLICATIONS

Textbooks

- **SooJean Han**. *Linear Systems: Theory and Application*. KAIST Research Series. Springer, Jan 2026

Academic Journals

- Published
 - * **SooJean Han**, Soon-Jo Chung, and John C. Doyle. Predictive Control of Linear Discrete-Time Markovian Jump Systems by Learning Recurrent Patterns. *Automatica*, 156, May 2023. [Link]
 - * **SooJean Han** and Soon-Jo Chung. Incremental Nonlinear Stability Analysis for Stochastic Systems Perturbed by Lévy Noise. *International Journal of Robust and Nonlinear Control*, 32(12):7174–7201, Aug 2022. [Link]
 - * Shih-Hao Tseng, **SooJean Han**, and Adam Wierman. Trading Throughput for Freshness: Freshness-Aware Traffic Engineering and In-Network Freshness Control. *ACM Transactions on Modeling and Performance Evaluation of Computing Systems (ToMPECS)*, 8(4):1–26, Mar 2023. [Link]
 - * Mo Chen, Sylvia L. Herbert, Haimin Hu, Ye Pu, Jaime Fernández Fisac, Somil Bansal, **SooJean Han**, and Claire J. Tomlin. FaSTrack: A Modular Framework for Real-Time Motion Planning and Guaranteed Safe Tracking. *IEEE Transactions on Automatic Control*, 66(12):5861–5876, Dec 2021. [Link]
- Preprints
 - * Soyeon Koo, **SooJean Han**, and Hyo-Sang Shin. Convex Reformulation of Nonlinear Joint Chance-Constrained Control via Spectral Iterative Risk Allocation. *IEEE Control Systems Letters*, Mar 2026. Under review
 - * Jungyo Jung and **SooJean Han**. Hierarchical Robust Control of Winch-Tethered Quadrotors via Adaptive Bias Compensation and Distance-Based Winch Regulation. *IEEE Control Systems Letters*, Mar 2026. Under review

Conferences

- Published
 - * Jeongyong Yang, KwangBin Lee, and **SooJean Han**. Heterogeneous Predictor-based Risk-Aware Planning with Conformal Prediction in Dense, Uncertain Environments. In *Neuro-Symbolic Systems (NeuS)*, Apr 2026. [Link]
 - * Minseok Jeong, **SooJean Han**, and Hyo-Sang Shin. Scalable Infinitesimal Generator-Based Koopman Learning for Long-Horizon Prediction. In *Learning for Dynamics and Control (L4DC) 2026*, Feb 2026. To appear
 - * Jeongyong Yang, Seunghwan Jang, and **SooJean Han**. SafeFlowMatcher: Safe and Fast Planning using Flow Matching with Control Barrier Functions. In *International Conference on Learning Representations (ICLR) 2026*, Feb 2026. To appear
 - * Eunwoo Sung and **SooJean Han**. Distributed Kalman-IMM Cooperative Estimator Synthesis for Large-Scale Networked Switching Systems. In *IEEE 64th Annual Conference on Decision and Control (CDC)*, Dec 2025
 - * **SooJean Han**, Minwoo Kim, and Ieun Choo. A Stochastic Robust Adaptive Systems Level Approach to Stabilizing Large-Scale Uncertain Markovian Jump Linear Systems. In *IEEE 63rd Annual Conference on Decision and Control (CDC)*, Sep 2024
 - * Mukul Chodhary, Kevin Octavian, and **SooJean Han**. Efficient Replay Memory Architectures in Multi-Agent Reinforcement Learning for Network Congestion Control. In *27th IEEE International Conference on Intelligent Transportation Systems (ITSC)*, Jul 2024

- * **SooJean Han**, Soon-Jo Chung, and Johanna Gustafson. Congestion Control of Vehicle Traffic Networks by Learning Structural and Temporal Patterns. In *Proceedings of The 5th Annual Learning for Dynamics and Control Conference*, volume 211 of *Proceedings of Machine Learning Research*, pages 903–914. PMLR, Jun 2023. [Link]
- * **SooJean Han**. Localized Learning of Robust Controllers for Networked Systems with Dynamic Topology. In *Proceedings of the 2nd Conference on Learning for Dynamics and Control (L4DC)*, volume 120 of *Proceedings of Machine Learning Research*, pages 687–696. PMLR, Jun 2020. [Link]
- * Shih-Hao Tseng, Carmen Amo Alonso, and **SooJean Han**. System Level Synthesis via Dynamic Programming. In *IEEE 59th Annual Conference on Decision and Control*, pages 1718–1725, Dec 2020. [Link]
- * Sylvia L. Herbert, Mo Chen, **SooJean Han**, Somil Bansal, Jaime F. Fisac, and Claire J. Tomlin. FaSTrack: A Modular Framework for Fast and Guaranteed Safe Motion Planning. In *IEEE 56th Annual Conference on Decision and Control (CDC)*, pages 1517–1522, Dec 2017. [Link]
- Preprints
 - * Eunwoo Sung, **SooJean Han**, and Hyo-Sang Shin. Localized Markov Jump Quadratic Optimal Control: A System-Level Approach. In *IEEE Conference on Decision and Control (CDC)*, Mar 2026. Under review
 - * Jeonghyeon Noh and **SooJean Han**. Communication Architecture Co-Design for Distributed Control via System Level Synthesis. In *IEEE Conference on Decision and Control (CDC)*, Mar 2026. Under review
 - * Nakgyu Yang, KwangBin Lee, Seunghwan Jang, and **SooJean Han**. Graph-Guided Safe Diffuser: Topological Graph Guidance for Safe Diffusion Planning. In *International Conference on Machine Learning (ICML)*, Jan 2026. Under review
 - * Minseok Jeong, Yechan Lee, Hyewon Choi, Jeongyong Yang, , and **SooJean Han**. A Gaussian Process Perspective on Overgeneralization in Random Network Distillation. In *International Conference on Machine Learning (ICML)*, Jan 2026. Under review

INVITED SEMINARS AND SYMPOSIUMS

Seoul National University, Graduate School of Dynamics/Control, Invited Talk	Dec 2025
Sungkyunkwan University, Department of Intelligent Robotics, Invited Talk	Oct 2025
International Conference for Women in Science and Technology (BIEN)	Aug 2025
Outstanding Young Faculty Seminar, ICROS	Jun 2025
KAIST Cho Chun Shik Graduate School of Mobility Invited Seminar	Feb 2025
Institute of Control Robotics and Systems (ICROS) Invited Talk	Dec 2024
Korea Robotics Conference (KRoC) New Faculty Seminar	Feb 2024
U. Michigan Ann-Arbor, Department of Mechanical Engineering, Invited Talk	Apr 2023
Stanford University, Department of Aeronautics and Astronautics, Invited Talk	Mar 2023
U. Pennsylvania, Department of MEAM, Invited Talk	Mar 2023
KAIST, School of Electrical Engineering, Invited Talk	Oct 2022

TEACHING & MENTORING EXPERIENCE

Courses Taught at KAIST			Aug 2023 - present
Semester	Course Number	Title	Class Size
Spring 2026	EE202	Signals and Systems	85
Fall 2025	EE899	Stochastic Control	40 (self-created topics course)
Spring 2025	EE210	Probability and Random Processes	70
Fall 2024	EE202	Signals and Systems	65
Spring 2024	EE581	Linear Systems	60
Fall 2023	EE488F	Learning Patterns for Autonomous Control	12 (self-created topics course)

Summer Undergraduate Research Fellow (SURF) Mentor

Jun 2022 - Sep 2022

Mentored one SURF student in the Autonomous Robotics and Control Lab (ARCL) at Caltech. Project was about congestion control of vehicle traffic intersection networks via adaptive signals. Relevant paper [here], presented at L4DC'23).

Teaching Assistant for Optimal Control and State-Estimation

Jan 2020 - Mar 2020

TA for CDS 112/Ae103a taught at Caltech during the Winter 2020 quarter. Provided lecture recitations and guest lectures on discrete and continuous-time Kalman filtering, particle filtering, the HJB equation, Euler-Lagrange equations, and the Pontryagin Maximum Principle. Created lecture notes for the course which can be found [here] and [here] (this one was made with students).

Summer Undergraduate Research Fellow (SURF) Mentor

Jun 2017 - Aug 2017

Mentored two SURF students in the Autonomous Robotics and Control Lab (ARCL) at Caltech. Project was about synchronization and formation-flying for multi-agent networks of Crazyflie quadrotors.

FELLOWSHIPS**NSF GRFP**

Jun. 2019 - Jun. 2022

Caltech Special EAS Fellowship

Jun. 2017 - Jun. 2019

TECHNICAL SKILLS**Software:** Python, MATLAB, C/C++, Java, LaTeX, Metapost, HTML/CSS, ROS**Version Control Tools:** Github, Bitbucket**Libraries/Toolboxes:** CVX, OpenAI Gymnasium, Simulink, PyTorch